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Serum adiponectin and soluble intercellular adhesion molecule-1 in patients with diabetic foot ulcers: a retrospective cohort study on the evaluation of disease severity and prognosis

Juan Wu[1,2], Ying Lu[3] and Ke Chen[1,2*]

Abstract

Objective This study assessed serum Adiponectin (APN) and Soluble Intercellular Adhesion Molecule-1 (sICAM-1) levels in diabetic foot ulcer (DFU) patients for evaluating disease severity and prognosis.

Methods In this retrospective cohort study, 92 DFU patients were assessed during a 6-month outcome evaluation period and categorized by ulcer healing into good prognosis (healed, $n=50$) or poor prognosis (unhealed, $n=42$). Serum APN and sICAM-1 levels were measured by Enzyme-Linked Immunosorbent Assay (ELISA) and and analyzed for correlations with disease severity and infection grades. Receiver Operating Characteristic (ROC) curves evaluated their predictive value for poor prognosis.

Results Serum APN was negatively correlated with infection and disease severity, whereas sICAM-1 exhibited a positive correlation. The good prognosis group had significantly higher APN and lower sICAM-1 levels compared to the poor prognosis group. Both biomarkers predicted poor prognosis: APN (Area Under the Curve (AUC) = 0.765), sICAM-1 (AUC = 0.841). Combining both markers yielded an AUC of 0.853.

Conclusion Serum APN and sICAM-1 levels are associated with DFU severity and prognosis, supporting their potential as predictive biomarkers in clinical practice.

Clinical trial number Not applicable.

Keywords APN, sICAM-1, Diabetic foot ulcers, Prognosis

*Correspondence:
Ke Chen
ckecsu@163.com
[1]Department of Geriatrics, Xiangya Hospital, National Clinical Research Center for Geriatric Disorders, Central South University, No.87, Xiangya Road, Kaifu District, Changsha, Hunan Province 410008, P.R. China
[2]National Clinical Research Center for Geriatric Disease (Xiangya Hospital), Central South University, Changsha 410008, China
[3]Department of General Medicine, Xiangya Hospital, Central South University, Changsha 410008, China

Introduction

Diabetes mellitus (DM) is an endocrine disorder characterized by persistent hyperglycemia and represents one of the most prevalent and rapidly growing diseases worldwide [1]. While current therapies can manage blood glucose levels, they do not offer a cure. Due to poor control, DM can lead to chronic complications, with various complications occurring even at the initial diagnosis of diabetes. Diabetic foot ulcers (DFU) are a frequently occurring chronic complication characterized by lower limb vascular obstruction, persistent foot infections, ulcers, and deep tissue damage [2]. It is estimated that globally, 537 million people have diabetes, with 19% to 34% of them developing DFU, and approximately 20% of DFU patients requiring lower limb amputation [3]. Currently, DFU treatment typically involves wound debridement, infection control, glucose regulation, and vascular intervention, with treatment costs approaching the total expenses of other diabetes complications [4]. However, DFU still exhibit poor healing, leading to amputation or death. Early risk stratification of DFU patients could potentially reduce hospitalization rates, disability rates, and mortality.

Intercellular Adhesion Molecule-1 (ICAM-1) is a protein belonging to the immunoglobulin superfamily synthesized by endothelial cells. Its primary function is to recruit and transport white blood cells through interactions with integrins expressed on white blood cells [5]. ICAM-1's unique structural domains allow it to serve as a biosensor, engaging with the cytoskeletal actin upon ligand binding and facilitating signal transduction [6]. In diseases associated with diabetes, ICAM-1 is found to be highly expressed in the patient's body and is believed to be involved in the development of these diseases [7, 8]. Adiponectin (APN) occurs in human circulating plasma at concentrations of 3–30 μg/ml and is encoded by the apM1 gene of human chromosome 3q27, which consists of three exons and two introns. The relative molecular weight of adiponectin is 30,000 (30 kDa), composed of 244 amino acids, including an amino terminal secretion signal sequence (aa 1–18), a specific sequence (aa 19–41), a collagen repeat sequence composed of 22 amino acids (aa 42–107), and a globular sequence (aa 108–244). The globular region is the key site of adiponectin bioactivity. APN plays a crucial role in the progression of diabetes, affecting insulin sensitivity, vascular inflammation, and atherosclerosis [9]. The role of APN in diabetic patients remains controversial. Previous studies have suggested that higher APN levels are associated with a lower risk of DM and have a cardioprotective effect [10]. However, conflicting results exist, with research indicating a positive correlation between serum and aqueous humor APN levels and the occurrence and progression of diabetic retinopathy [11].

However, the combined prognostic significance of serum APN and soluble Intercellular Adhesion Molecule-1 (sICAM-1) has not been systematically examined in DFU, a condition where both metabolic disturbance and sustained inflammation converge. Current prognostic tools primarily rely on clinical staging (e.g., Wagner classification) and may not fully reflect the underlying systemic pathophysiology. Therefore, there is a need for reliable and measurable serum biomarkers to improve early risk stratification. This study aimed to evaluate whether serum levels of APN and sICAM-1, individually and in combination, could predict poor prognosis in DFU patients. The primary outcome was ulcer healing status at 6-month outcome assessment period (categorized as good prognosis: healed; poor prognosis: unhealed). Secondary outcomes included the correlation of these biomarkers with DFU severity (Wagner grade) and infection severity. Specifically, we investigated serum concentrations of APN and sICAM-1, analyzed their relationships with ulcer severity and infection status, and assessed their combined predictive ability for adverse outcomes. The findings may provide evidence to support the potential utility of this dual-biomarker panel in the clinical assessment of DFU.

Subjects and methods

Study population

This study was designed as a retrospective cohort study. 92 DFU patients admitted to our hospital from January 2023 to January 2024 were included in the analysis. Patient outcomes (ulcer healing status) were assessed through a review of medical records at 6 months after the initial assessment, with 50 cases in the good prognosis group (ulcer healed) and 42 cases in the poor prognosis group (ulcer not healed). Inclusion criteria followed the "Chinese Guideline on Prevention and Management of Diabetic Foot (2019 edition)" and the diagnostic criteria for type 2 diabetes (T2DM) [12, 13]. Exclusion criteria included severe cardiovascular, cerebrovascular, pulmonary diseases, mental disorders, autoimmune diseases, malignant tumors, recent major surgeries, ulcer malignancy, type 1 diabetes, other diabetic complications, other infectious diseases, osteomyelitis and long-term use of steroids or immunosuppressants. The study was approved by the hospital's ethics committee (protocol: 2024-166). Given the retrospective design and the use of anonymized residual serum samples originally obtained for routine clinical care during the patients' hospitalization, the requirement for written informed consent was waived by the ethics committee. All procedures adhered to Helsinki Declaration principles and institutional regulations.

Methods

Treatment

Patients were treated with insulin to control blood glucose levels and received comprehensive treatments, including anti-infection measures, circulatory support, nutritional support, and wound management. All subjects had the necrotic tissue removed from the diabetic foot wound, and the dressing was changed every other day. The patient should not go to the ground and should not put pressure on the wound. During the 6-month outcome assessment period, patients received ongoing care at their local community hospital or primary care clinic.

Assessment of disease severity and infection level

Disease severity and infection level were assessed based on the final clinical evaluation recorded during the 6-month outcome assessment period. Disease severity was graded according to the "Wagner Classification for DFU," [14] which has five levels: 1 for mild, 2–3 for moderate, and 4–5 for severe. The severity of DFU infection was classified according to the "Infectious Diseases Society of America Clinical Practice Guideline for the Diagnosis and Treatment of Diabetic Foot Infections," [15] with levels 1 for no infection, 2 for mild infection, 3 for moderate infection, and 4 for severe infection .

Laboratory testing

Peripheral blood (5 mL) was collected from each subject during the final clinical evaluation within the 6-month outcome assessment period. Samples were centrifuged at 1,000×g for 15 min at 4 °C. Routine laboratory measurements were accessed and the residual supernatant was stored at -70 °C until analysis. Enzyme-Linked Immunosorbent Assay (ELISA) kits purchased from Shanghai Yubo Biotechnology Company were used to measure APN and sICAM-1 levels in serum samples, following the provided protocols. Absorbance was measured at 450 nm using a microplate reader (Thermo Multiskan FC, USA).

Statistical analysis

A post-hoc power analysis (G*Power 3.1, $\alpha = 0.05$, effect size $d = 0.6$) indicated a power of 0.81 for the total sample of 92 patients (42 in the poor prognosis group and 50 in the poor prognosis group), exceeding the 0.80 threshold, supporting adequate sample size for detecting clinically meaningful differences. Statistical analysis was performed using SPSS 21.0 (IBM Corp, Armonk, NY, USA), and GraphPad Prism 9.5.0 was used for data visualization. Normality of continuous data was assessed using the Shapiro-Wilk test. Normally distributed continuous data are presented as mean ± standard deviation ($\bar{x} \pm s$): comparisons between two groups were performed using independent samples t-tests, and comparisons among multiple groups were performed using one-way analysis of variance (ANOVA), with post-hoc analysis conducted using Tukey's multiple comparisons test. Non-normally distributed continuous data are presented as median (25th to 75th percentile): comparisons between two groups were performed using the Mann-Whitney U test, and comparisons among multiple groups were performed using the Kruskal-Wallis test. Categorical variables were compared using the Chi-square test. Correlations between variables were evaluated using Pearson correlation analysis for normally distributed data and Spearman's rank correlation analysis for non-normally distributed data. The receiver operating characteristic (ROC) curves were used to evaluate the diagnostic value of serum APN and sICAM-1 for predicting poor prognosis in DFU patients. A multivariable Cox proportional hazards regression model was constructed to identify independent factors associated with prognosis. To control for confounding, variables showing significant associations in univariate analyses ($P < 0.05$) and those of clinical relevance were included as covariates in the multivariable model. Multicollinearity among covariates was assessed using the variance inflation factor (VIF), with VIF < 5 considered acceptable, indicating no substantial multicollinearity. A two-tailed P value < 0.05 was considered statistically significant.

Results

Comparison of clinical data between poor prognosis and good prognosis groups

The study flow chart is shown in Fig. 1. The poor prognosis group showed significantly higher values for DFU duration, fasting blood glucose, glycated hemoglobin, Wagner grade, and infection severity grade compared to the good prognosis group, with all differences being statistically significant (all $P < 0.05$). There was no significant difference in gender, age, BMI, ulcer area and ulcer location between the two groups of diabetes foot ($P > 0.05$). See Table 1.

Serum concentrations of APN and sICAM-1 in DFU patients with different infection severity

Compared with the patients with grade 1 to 2 infection severity, the serum APN level in patients with grade 3 to 4 infection severity was significantly decreased, and the sICAM-1 level was significantly increased, with statistical significance (all $P < 0.001$). See Fig. 2. Spearman rank correlation analysis confirmed a significant negative correlation between serum APN concentrations and infection severity ($r = -0.894$, $P < 0.001$), and a significant positive correlation between serum sICAM-1 concentrations and infection severity ($r = 0.890$, $P < 0.001$).

Table 1 Comparison of clinical data between poor prognosis and good prognosis groups ([n(%)], $\bar{x} \pm s$)

Group	Poor prognosis group ($n=42$)	Good prognosis group ($n=50$)	χ^2/t	*P*
Gender			1.104	0.293
Male	25(59.52)	35(70.00)		
Female	17(40.48)	15(30.00)		
Age (years)	63.57 ± 12.10	61.36 ± 12.58	0.854	0.395
BMI (kg/m^2)	22.78 ± 2.56	22.96 ± 2.83	0.318	0.752
DFU duration (weeks)	8.14 ± 3.84	4.36 ± 2.23	5.883	<0.001
Ulcer area (cm^2)	7.55 ± 3.59	6.58 ± 3.20	1.370	0.174
White blood cell count (×10^9/L)	11.79 ± 2.71	10.84 ± 2.61	1.709	0.091
Fasting blood glucose (mmol/L)	10.80 ± 1.81	8.86 ± 2.05	4.767	<0.001
Glycated hemoglobin (%)	9.20 ± 1.92	7.34 ± 1.53	5.171	<0.001
Ulcer location				
Instep	3(7.14)	2(4.00)	1.26	0.965
Ankles	5(11.90)	4(8.00)	1.64	0.799
Foot side	4(9.52)	3(6.00)	1.19	0.256
Toes	11(26.19)	14(28.00)	3.78	0.324
Sole of feet	19(45.24)	27(54.00)	5.47	0.071
Wagner grade			14.571	<0.001
Grade 1	3(7.14)	23(46.00)		
Grade 2 ~ 3	15(35.71)	20(40.00)		
Grade 4 ~ 5	24(57.14)	7(14.00)		
Infection severity grade			8.938	0.003
Grade 1 ~ 2	7(16.67)	23(46.00)		
Grade 3 ~ 4	35(83.33)	27(54.00)		

Note: BMI, body mass index; DFU, diabetic foot ulcer. Continuous variables are presented as mean ± standard deviation and compared using independent sample t-test. Categorical variables are presented as n (%) and compared using the Chi-square test. A P value < 0.05 was considered statistically significant

Serum concentrations of APN and sICAM-1 in DFU patients with different disease severity

As Wagner grade increased, the serum levels of APN gradually decreased, while the levels of sICAM-1 gradually increased (4 ~ 5 grade > 2 ~ 3 grade > 1 grade), with statistically significant differences (all $P < 0.001$). See Fig. 3. Spearman rank correlation analysis confirmed a significant negative correlation between serum APN concentrations and disease severity (r=-0.914, $P < 0.001$), and a significant positive correlation between serum sICAM-1 concentrations and disease severity ($r = 0.958$, $P < 0.001$).

Serum concentrations of APN and sICAM-1 in DFU patients with different prognoses

Compared with patients with unfavorable prognosis, those with favorable prognosis exhibited a significant increase in serum APN levels and a marked decrease in sICAM-1 levels, with statistically significant differences (both $P < 0.001$). See Fig. 4.

Predictive value of serum APN and sICAM-1 for poor prognosis in DFU patients

The ROC curve analysis revealed that serum APN and sICAM-1 had a certain predictive value for adverse prognosis in DFU patients, with AUC values of 0.765 and 0.841, respectively. The combined diagnostic efficacy of both was represented by an AUC of 0.853. See Table 2; Fig. 5.

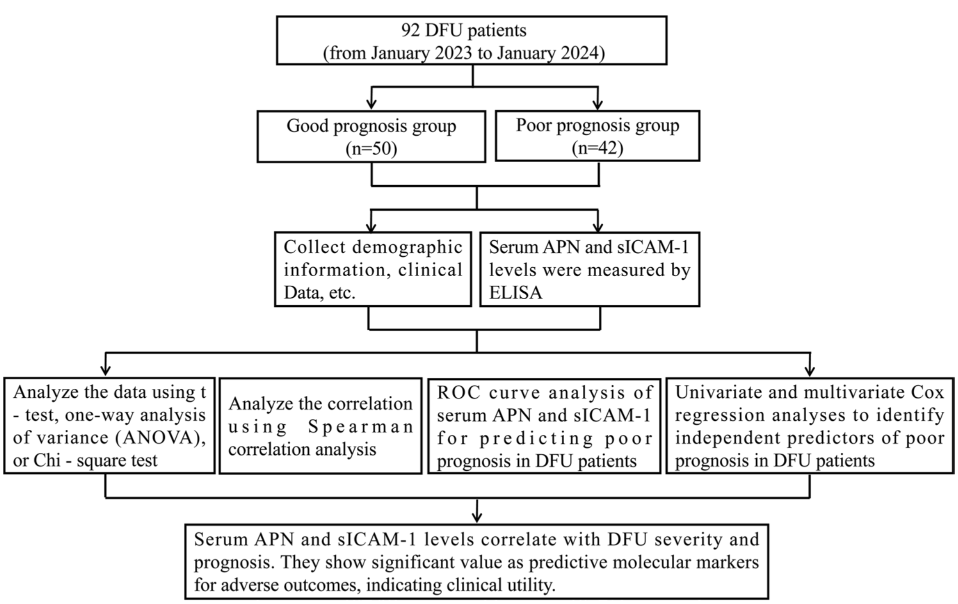

Fig. 1 Flowchart of patient enrollment and study design for DFU patients

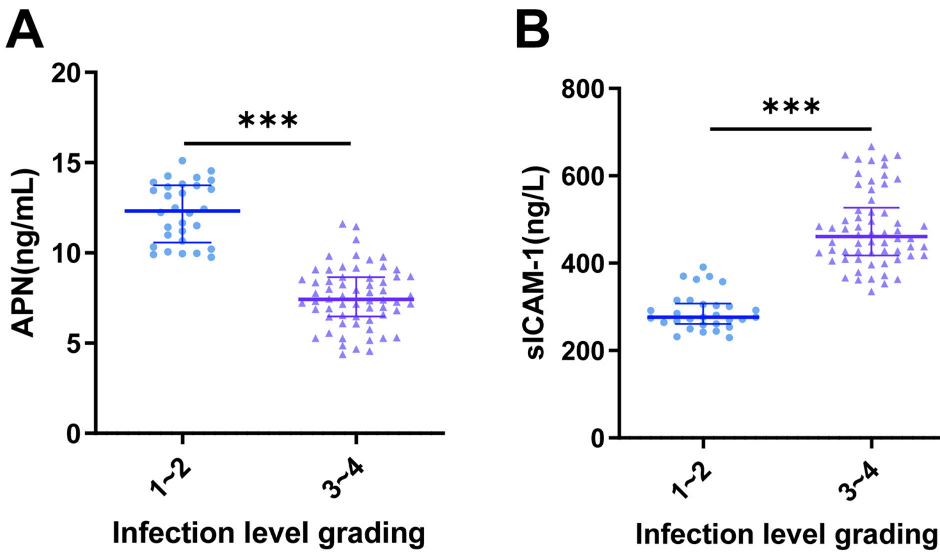

Fig. 2 Serum concentrations of APN and sICAM-1 in DFU patients with different infection severity. **A**: Serum APN levels in patients with different infection severity. **B**: Serum sICAM-1 levels in patients with different infection severity. Note: DFU = Diabetic Foot Ulcer; APN = Adiponectin; sICAM-1 = Soluble Intercellular Adhesion Molecule-1. Differences between two independent groups were compared using the independent samples t-test. The level of statistical significance was set at *P* < 0.05. *** represents *P*<0.001

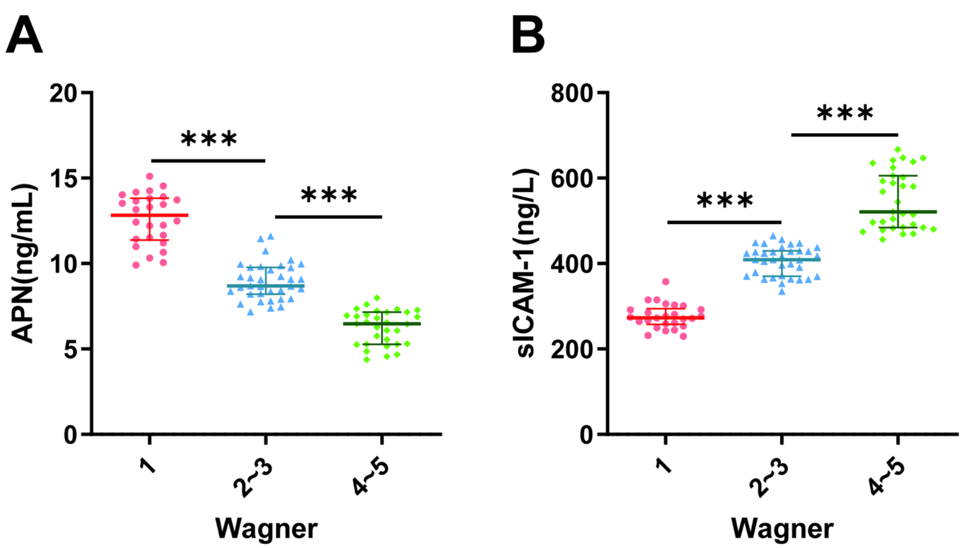

Fig. 3 Serum concentrations of APN and sICAM-1 in DFU patients with different disease severity. **A**: Serum APN levels in patients with different disease severity. **B**: Serum sICAM-1 levels in patients with different disease severity. Note: DFU = Diabetic Foot Ulcer; APN = Adiponectin; sICAM-1 = Soluble Intercellular Adhesion Molecule-1. Differences among multiple groups were compared using one-way analysis of variance (ANOVA) followed by Tukey's multiple comparisons test. The level of statistical significance was set at *P* < 0.05. *** represents *P*<0.001

Univariate and multivariable Cox proportional hazards regression analysis

Univariate and multivariable Cox proportional hazards regression analyses were performed to identify independent predictors of poor prognosis in DFU patients, with the 6-month ulcer healing status as the dependent variable (good prognosis: healed; poor prognosis: unhealed). Univariate Cox regression analysis was initially conducted with the following covariates from Table 1: gender, age, BMI, DFU duration, ulcer area, white blood cell count, fasting blood glucose, glycated hemoglobin, ulcer location, Wagner grade, infection severity grade, and serum levels of APN and sICAM-1. The results of the univariate analysis revealed that DFU duration, fasting blood glucose, glycated hemoglobin, Wagner grade, infection severity grade, and serum levels of APN and sICAM-1 were significantly associated with poor prognosis (all *P* < 0.05; see Table 3).

Subsequently, variables that showed statistical significance in the univariate analysis were included in

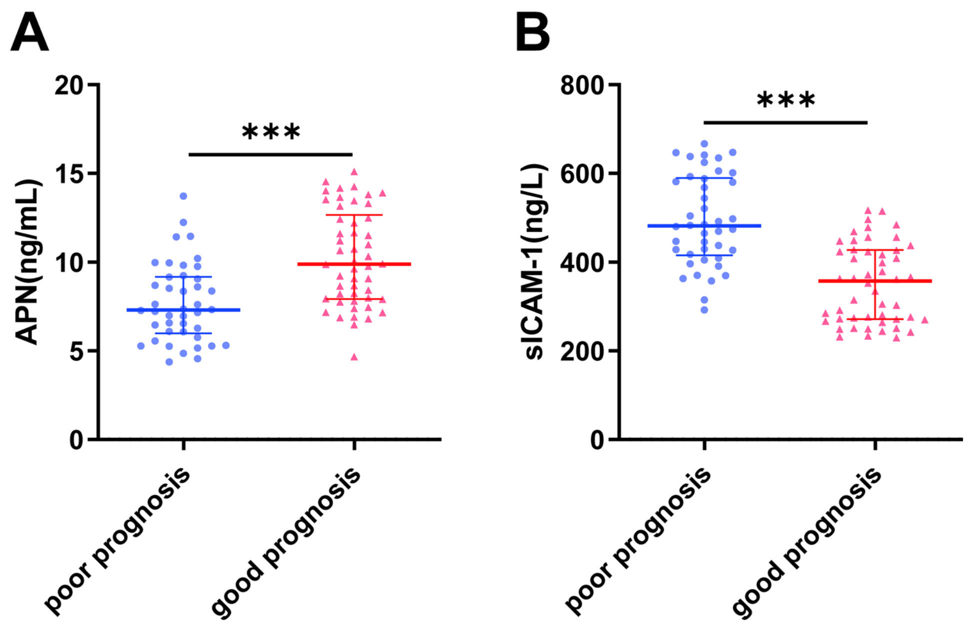

Fig. 4 Serum concentrations of APN and sICAM-1 in DFU patients with different prognoses. **A**: Serum APN levels in DFU patients with different prognoses. **B**: Serum sICAM-1 levels in DFU patients with different prognoses. Note: APN = Adiponectin; sICAM-1 = Soluble Intercellular Adhesion Molecule-1. Differences between two independent groups were compared using the independent samples t-test. The level of statistical significance was set at $P < 0.05$. *** represents $P<0.001$

Table 2 Predictive value of serum APN and sICAM-1 for poor prognosis in DFU patients

Diagnostic Index	AUC	95%CI	Standard Error	Cut-off	Sensitivity (%)	Specificity (%)	P
APN	0.765	0.669–0.861	0.048	7.79 (ng/mL)	82.00	59.52	< 0.001
sICAM-1	0.841	0.763–0.918	0.039	427.22 (ng/L)	76.00	73.81	< 0.001
Combined	0.853	0.778–0.928	0.038	-	84.00	72.09	< 0.001

Note: The predictive value of serum APN and sICAM-1 for poor prognosis was evaluated using ROC curve analysis. AUC = Area under the ROC curve, 95%CI = 95% confidence interval. A P value < 0.05 was considered statistically significant

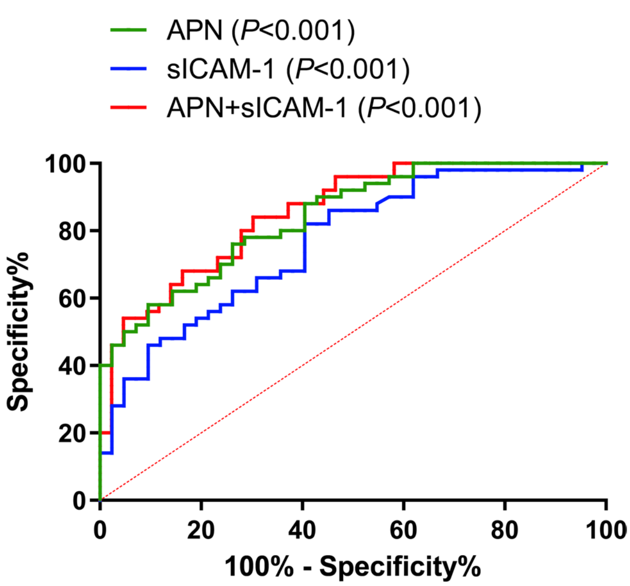

Fig. 5 ROC curves of serum APN, sICAM-1, and their combination in predicting poor prognosis in DFU patients

the multivariable Cox regression model. The multivariable analysis demonstrated that fasting blood glucose (HR = 1.185, 95% CI: 1.001–1.402, $P = 0.048$), APN (HR = 1.554, 95% CI: 1.012–2.385, $P = 0.044$), and sICAM-1 (HR = 1.014, 95% CI: 1.003–1.026, $P = 0.012$) were independent risk factors for poor prognosis in DFU patients (Table 3).

Discussion

The findings of this study demonstrate that serum levels of APN and sICAM-1 are significantly associated with infection severity, disease severity (Wagner grade), and clinical prognosis in patients with DFU. Specifically, lower APN and higher sICAM-1 levels correlate with more severe infection, higher Wagner grade, and poorer healing outcomes at 6 months. ROC analysis further supports their combined utility as prognostic biomarkers.

Our results regarding APN are consistent with the established biological functions of this adipokine and previous clinical observations. Adiponectin is well-recognized for its anti-inflammatory, insulin-sensitizing, and anti-atherosclerotic properties [16, 17]. Abdalla MMI et

Table 3 Univariate and multivariable Cox proportional hazards regression analyses for prognostic factors in DFU patients

Variable	Univariable		Multivariable	
	HR (95% CI)	P	HR (95% CI)	P
Gender	0.605(0.325–1.129)	0.115	/	/
Age	1.009(0.984–1.033)	0.490	/	/
BMI	1.019(0.921–1.127)	0.714	/	/
DFU duration	1.176(1.088–1.271)	< 0.001	0.912(0.784–1.060)	0.229
Ulcer area	1.081(0.988–1.182)	0.092	/	/
White blood cell count	1.094(0.977–1.226)	0.120	/	/
Fasting blood glucose	1.283(1.095–1.502)	0.002	1.185(1.001–1.402)	0.048
Glycated hemoglobin	1.285(1.129–1.463)	< 0.001	1.104(0.943–1.292)	0.220
Ulcer location	0.934(0.734–1.189)	0.580	/	/
Wagner grade	1.684(1.330–2.134)	< 0.001	1.157(0.410–3.264)	0.783
Infection severity grade	2.058(1.406–3.012)	< 0.001	1.186(0.468–3.008)	0.719
APN	0.782(0.685–0.894)	< 0.001	1.554(1.012–2.385)	0.044
sICAM-1	1.007(1.004–1.009)	< 0.001	1.014(1.003–1.026)	0.012

Note: Multivariable Cox proportional hazards regression analysis was performed to identify independent predictors of poor prognosis in DFU patients (defined as ulcer non-healing within 6 months). The dependent variable was prognosis status (good prognosis: healed; poor prognosis: unhealed). HR = hazard ratio, 95% CI = 95% confidence interval. A P value < 0.05 was considered statistically significant

al., in a comprehensive review, highlighted that adiponectin can inhibit pro-inflammatory cytokine production, increase the secretion level of vascular endothelial growth factor, and is involved in glucose metabolism, immune regulation, and extracellular matrix remodeling, suggesting a negative association with DFU severity [18]. The significant decrease in serum APN observed in our patients with severe infection/Wagner grade and poor prognosis aligns with this protective role. It suggests that a relative deficiency of this beneficial adipokine may contribute to impaired healing in DFU, possibly through exacerbated inflammation and metabolic dysregulation.

Conversely, our finding of elevated sICAM-1 levels in severe and poorly prognostic DFU patients aligns with its role as a key mediator of endothelial activation and leukocyte recruitment during sustained inflammation. ICAM-1 levels is upregulated in diabetic conditions, contributing to microvascular complications [7, 19]. Kang HJ et al. demonstrated in a diabetic mouse wound model that accelerated healing following a therapeutic intervention was associated with decreased levels of the pro-inflammatory marker ICAM-1 [20]. This inverse relationship between wound healing and ICAM-1 levels directly supports our clinical observation that higher circulating sICAM-1 levels are associated with poor ulcer healing. Furthermore, meta-analytic evidence confirms that elevated circulating adhesion molecules, including sICAM-1, increase the risk of type 2 diabetes and its complications [21]. Thus, our data corroborate the concept that sICAM-1 is a marker of persistent inflammatory and endothelial dysfunction hindering DFU resolution.

The novel aspect of our study lies in evaluating the combined prognostic value of these two biomarkers, which reflect different but interconnected pathophysiological axes—metabolic regulation (APN) and inflammatory endothelial activation (sICAM-1). While individual AUCs for APN (0.765) and sICAM-1 (0.841) showed good predictive ability, their combination yielded a slightly higher AUC (0.853). This suggests that assessing both markers may provide a more comprehensive risk stratification than either alone, potentially capturing a broader spectrum of the systemic pathophysiology underlying DFU progression. This combined approach may offer an advantage over traditional staging systems like the Wagner classification by adding objective, serum-based molecular information.

Our multivariable Cox regression identified APN and sICAM-1, along with fasting blood glucose, as independent risk factors for poor prognosis, reinforcing their potential clinical relevance. However, several limitations must be acknowledged. First, this was a single-center retrospective study with a modest sample size, which may limit the generalizability of our findings and introduce selection bias. Second, although we controlled for several important clinical and demographic variables through multivariable Cox regression and assessed multicollinearity, residual confounding from unmeasured or unadjusted factors (such as specific antibiotic regimens, detailed nutritional status, or psychosocial factors) cannot be excluded. The sample size also limits the stability of the regression estimates and the power to detect smaller but potentially meaningful associations. Third, we measured biomarker levels at a single time point (6-month outcome period). Without serial measurements during treatment, we cannot determine their dynamic changes or utility in monitoring therapeutic efficacy. Future prospective studies that systematically document treatment details (e.g., types of antibiotics, wound care techniques) and incorporate longitudinal biomarker monitoring would help clarify the relationship between therapeutic interventions and changes in these markers. Future prospective, multicenter studies with larger cohorts and longitudinal sampling are warranted to validate these biomarkers and establish their clinical application thresholds. Additionally, exploring the mechanisms linking APN and sICAM-1 to specific cellular processes in DFU wounds could provide deeper insights.

From a translational perspective, the measurement of APN and sICAM-1 in patients with DFU offers a promising avenue for enhancing individualized prognosis and management. In this particularly vulnerable patient population—often characterized by multiple comorbidities, impaired healing, and high risk of amputation—the availability of reliable, minimally invasive biomarkers is of paramount importance. Our study underscores that APN and sICAM-1 are not merely research markers but hold tangible clinical potential: they may aid in the early triage of high-risk ulcers, inform decisions regarding the intensity of systemic anti-inflammatory or metabolic interventions, and serve as objective indicators of treatment response. The novelty of our work resides in the integrative use of these two biomarkers—each representing a distinct yet synergistic pathophysiological domain—to generate a composite prognostic signal. This approach moves beyond conventional staging by incorporating systemic, molecular insights into what has traditionally been a locally assessed condition.

The clinical translation of the combined APN/sICAM-1 panel may be advanced through four sequential stages: conducting multi-center prospective studies to standardize detection protocols and validate cutoff values; performing health-economic evaluations to demonstrate cost-effectiveness relative to conventional staging; piloting a dual-track assessment model that integrates biomarker profiling with clinical evaluation in regional referral centers; and developing rapid point-of-care testing technologies alongside targeted clinician training to enhance accessibility and interpretation. This structured implementation framework highlights the actionable steps needed to translate this biomarker panel from research evidence into routine clinical practice. If validated in larger, prospective cohorts, the APN–sICAM-1 panel could contribute to a more nuanced, mechanism-based stratification of DFU patients, ultimately supporting timely and tailored clinical actions aimed at reducing disability and improving outcomes.

Conclusion

This study confirms that serum APN and sICAM-1 levels are strongly correlated with the severity and prognosis of DFU. The findings are in agreement with existing literature on their respective roles in metabolism and inflammation, with lower APN and higher sICAM-1 reflecting more severe disease states and poorer healing outcomes. The combination of these two biomarkers shows promise as a supplementary tool for prognostic assessment, potentially aiding in the early identification of high-risk patients who may benefit from more aggressive or targeted interventions. However, further validation in larger, prospective, and multi-center cohorts is required before clinical translation can be considered.

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Author contributions
Ke Chen. submitted ethics approval, collected the data. Juan Wu started the original draft of the manuscript. Juan Wu and Ying Lu was the supervising investigator; he prepared the original elements of the protocol and supervised the data collection. Ke Chen Reviewed the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval
The Ethics Committee of Xiangya Hospital of Central South University approved this investigation (Protocol: 2024-166), with informed consent requirements waived given the retrospective design. All procedures adhered to Helsinki Declaration principles and institutional regulations.

Consent to participate
Not applicable.

Consent to publication
Not applicable.

Competing interests
The authors declare no competing interests.

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